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Vinay Ram Gazula

Graduate Student at NJIT

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Aspiring data scientist with a solid foundation in mechanical engineering, seeking a role as a Summer Intern in the field of data science, to leverage an unique blend of analytical, engineering and programming skills.

Education

Master of Science, *New Jersey Institute of Technology* — **Sep 2023 - May 2025** | NJ, USA
Data Science Computational Track — GPA: 3.83
Relevant Coursework: Applied Statistics, Introduction to Big Data, Data Managment Systems Design, Advanced Database Systems Design, Machine Learning and Data Analytics with R Program

Bachelor of Technology, *SRM University AP* — **Aug 2017 - May 2021** | AP, India
Mechanical Engineering — GPA: 3.52
Awarded with merit scholarship of 50% tuition fee for the entire bachelor's degree

Skills

Programming Languages	Python, R, Bash, C/C++, Java, Pig, Hadoop, MATLAB	
Database Languages	SQL, MySQL, SQLite, PL/SQL, Oracle, BigQuery, MongoDB	
Data Analytics	Spreadsheets, Pandas, Tidyverse, Matplotlib, Plotly, Tableau	
Tools	**Git**	GitHub
	Cloud	AWS EC2
	IDE / Text Editor	VS Code, RStudio, Sublime Text, KATE, Vim
	Documentation	MS Office, LaTeX, R Markdown
	Operating Systems	Linux, MacOS, Windows, UNIX
Libraries	NumPy, SymPy, Tensorflow, Keras, PyTorch, yfinance, Streamlit, ggplot	
CAD & CFD Softwares	Solidworks 2018, CATIA V5, Tecplot 360, Gmsh, ANSYS Fluent, AutoCAD	

Experience

Research Assistant — **Oct 2023 - Present**
Under Dr. Jason T. L. Wang (Professor, Computer Science, NJIT) — Newark, USA

- Integrated Explainable AI (xAI) tools and leveraged interpretability into machine learning and deep learning tools used for space weather research in collaboration with NSF / NASA (`https://nature.njit.edu/solardb/`)
- Incorporated xAI tools LIME, SHAP, PDP and ALE plots into a transformer-based framework "SolarFlareNet", used to forecast whether a $\geq M$-class flare occurs within $24, 48$ & 72 hours using SHARP parameters
- Submitted a conference paper in 37th International FLAIRS Conference which is currently under review

CAD Designer — **Mar 2022 - Apr 2023**
Jayram Buildeng Solutions Pvt. Ltd. — Vijayawada, India

- Designing 2D CAD drawings using AutoCAD
- Producing high quality, accurate, and timely reports for clients
- Actively organised and participated in client meetings and contributed in promoting the services provided by the firm
- Acquired 3 new clients to the firm increasing it's turnover by 14%

Research Intern — **Sep 2021 - Feb 2022**
Under Dr. Satya P. Jammy (Associate Professor, Mechanical Department, SRM University AP) — Amaravati, India

- Worked on a Science Education and Research Board (SERB) govt. of India project titled "Wall Effects in Shock Wave Boundary Layer Interaction"
- Contributed in the development of solver code for gaseous reactions and viscous properties
- Validated the solver for different flow configurations like Single ramp, Double ramp, etc
- Published a conference paper in 26th National and 4th International ISHMT-ASTFE Heat and Mass Transfer Conference

Projects

Interpretable Machine Learning for Flare Prediction | ***Python, Tensorflow, ML, DL, xAI*** — **Oct 2023 - Ongoing**
Research Assistant — Project Source

- Integrated xAI tools like LIME, SHAP, PDP and ALE plots into multiple machine learning and deep learning tools used for predictiong space weather
- Pioneered the integration of interpretability techniques into a transformer-based machine learning model named "SolarFlareNet"
- Studied the feature importance and feature interactions between the SHARP magnetic parameters used to forecast solar flares for NOAA selected active regions

eComputer Store Database System | *MySQL, Python, Streamlit, Pandas* **Sep 2023 - Dec 2023**
Data Managment Systems Design Course Project Project Source

- Designed and developed a working database for an eComputerStore with all mentioned functionalities
- Followed a traditional approach for building database by designing an Enhanced Entity Relationship (EER) diagram, transforming the EER model to a Relational database schema and creating the database tables in MySQL using the Relational schema
- Developed a web application using Streamlit which connects to the MySQL server using a pipeline

MURPH Analytics | *Python, Streamlit, Pandas, Matplotlib, Plotly* **Mar 2023 - May 2023**
Personal Project Project Source

- Built an interactive dashboard using Streamlit to post-process and perform different analytical studies on the simulation data generated from the MURPH solver
- Implemented unstructured mesh plotting and 2D contour plots using Plotly

MURPH | *Python, SymPy, NumPy, OpenMP, CUDA, OP2-API, CFD* **Dec 2022 - Ongoing**
In collaboration with Dr. Satya P. Jammy (Associate Professor, Mechanical Department, SRM University AP) Project Source

- A Multi-dimensional (2-D/3-D), Unstructured-mesh and Reactions based Parallel solver for Hyper-sonic flow regimes (MURPH)
- New design approach is used to build the solver code so that the parallel code generation can be fully automated
- 2D Ideal gas solver code is developed and is validated using existing literature
- Work is underway for integrating species reactions and development of a 3D solver

AlgoTrade API | *Python, yfinance, Pandas, Tensorflow, ks-api-client* **May 2022 - Oct 2022**
Personal Project Project Source

- Fully automated NSE Stock/Equity trading-bot for Kotak Securities (KOTAK TRADE API) with integrated back testing and ML algorithms for price and trend predictions
- The bot gathers any desired stock's historical price data, performs technical analysis using technical indicators and places orders to the stock broker using the broker API

Accelerating a Hypersonic CO_2 Reaction Solver | *Python, SymPy, NumPy, OpenMP, CUDA, OP2-API, CFD* **Jul 2020 - May 2021**
B.Tech Final Year Project Project Source

- A 2-D unstructured finite volume method based solver capable of simulating hypersonic flows in CO_2 rich Martian atmosphere with chemical kinetics for eight species consisting of $N_2, O_2, NO, O, N, CO_2, CO$ and C incorporated in it is developed using pYthon
- The solver is parallelized to run on single or multiple CPUs and Nvidia GPUs in order to attain greater computational speeds using OP2 application programming interface

Research Publications

FLAIRS-37 **May 2024**
The Florida Artificial Intelligence Research Society (FLAIRS) In Review

- Vinay, G., Yasser A., Jason T., "An Interpretable Transformer Model for Operational Flare Forecasting"

IHMTC-2021 **Dec 2021**
ISHMT-ASTFE Heat and Mass Transfer Conference Publication

- Vinay, G., Dinesh, D., Satya, P., "Accelerating a Hypersonic CO_2 Reaction Solver", ***Proceedings of the 26th National and 4th International ISHMT-ASTFE Heat and Mass Transfer Conference December 17-20, 2021, IIT Madras, Chennai-600036, Tamil Nadu, India***

Certifications

Google Data Analytics Professional Certificate **Dec 2022 - Apr 2023**
Coursera Certificate

Algorithmic Trading & Quantitative Analysis Using Python **Mar 2023 - Apr 2023**
Udemy Certificate

Introduction to Programming with MATLAB **Apr 2020 - May 2020**
Coursera Certificate

Autodesk Fusion 360 Integrated CAD/CAM/CAE **Feb 2020 - Mar 2020**
Coursera Certificate

Robo Feast 2018 **Feb 2018**
Certificate of Participation
Participated in the 2 days National Level Workshop cum Championship on "*WiFi-Controlled Robotics*" organized by ***Revert Technology Pvt. Ltd. & EDC, IIT Roorkee***

March 11, 2024